

Rana Talukder

Netrakona, Mymensingh, Bangladesh

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Education

Bachelor of Science in Electrical and Electronics Engineering

March 2020 – June 2025

Chittagong University of Engineering and Technology (CUET)

- CGPA: 3.50 / 4.00 ◦ Duration: 4 years

Research Experience

Undergraduate Thesis

CUET

Supervisor: Prof. Dr. Mahmud Abdul Matin Bhuiyan

Dec 2024 – June 2025

- Thesis Title: **Performance Analysis and Current Matching Optimization of KGeCl₃ based Tandem Solar Cell using SCAPS-1D with a Machine Learning Approach.**
- Successfully developed an idea to predict the top cell thickness to match current by using ML.

Research Publications

Journal Articles

- **Rana Talukder a,1, Md. Adnan Faisal Siddique b,1, MD. Ashaduzzaman Ove a, Supreyo Chakravorty Pretom a, Mahmud Abdul Matin Bhuiyan a, Foyzul Karim a*.** **Performance Analysis and Current Matching Optimization of KGeCl₃ based Tandem Solar Cell using SCAPS-1D with a Machine learning Approach** In: *Next Material*(Accepted on 04/04/2026). DOI: <https://doi.org/10.1016/j.nxmte.2026.101978>
- **Rana Talukder a, Foyzul Karim a*, Supreyo Chakravorty Pretom a, K.M. Mehedi Hassan b, Debjyoti Banik b, Md. Habibur Rahman Aslam a.** **High-Efficiency Stable Ba₃PBr₃-Based Lead-Free Perovskite Solar Cells: Insights from DFT and SCAPS-1D Modeling** In: *Results in Materials* (Accepted on 29/06/2026). DOI: <https://doi.org/10.1016/j.rinma.2026.101000>
- Md. Habibur Rahman Aslam*, Foyzul Karim, **Rana Talukder**, Supreyo Chakravorty Pretom. **Numerical Investigation of CsPbI₂Br-based Perovskite Solar Cell Using Diverse ETL/HTL Combinations** In: *Jurnal of Chemistry* (Under Review).
- Supreyo Chakravorty Pretom, Foyzul Karim*, Hasibul Hasan, **Rana Talukder**, Md. Habibur Rahman Aslam. **A Novel Destructive Interference Method for THD Reduction in Seven-Level Cascaded H-Bridge Inverters** In: *Electric Power Systems Research* (Under Review).

Projects

Three-Digit Object Counter

- Designed a precise object counter.
- Tools: Arduino Uno, LCD

Distance Measurement of an object

- Capable of accurately measuring object distance.
- Tools: PIC Microcontroller, PICKit Programmer, Ultrasonic Sensor.

Electrical Design of 2300 sq. ft. Residential Building

- Fittings, Fixtures and Conduit drawing.
- Tools: Autocad (electrical).

Technical and Software Skills

Programming: C, C++, HTML, ML(Basic Level).

Software: SCAPS-1D, COMSOL Multiphysics, MATLAB, Cadence Virtuoso.

Typography: LaTeX, MS Word, PowerPoint.
Languages: Bengali (Native), English (Fluent).

Industrial Attachment

- Bangladesh Telecommunications Company Ltd.(**BTCL**) Transmission Exchange (POP), Nandankanan Agrabad, Chittagong. 2024
Gained observational exposure and learning how telecom. networks operate and how different services are delivered.
- Duration : 24 November - 12 December

Training and Workshops

- Power System Operation and Service Design** 2025
 - Conducted By: EDGE Project, ICT Division.
 - Contact Hours: 80 hrs.
- Solar System Design and Management** 2025
 - Conducted By: Daffodil International University, Ashulia, Dhaka.
 - Contact Hours: 32 hrs.

Industrial Visits

- General Electric Manufacturing Company Limited (**GEMCO**), Chittagong 2024
A Bangladesh government owned company that manufactures electric transformers.
- Chittagong Thermal Power Plant, Raozan, Chittagong 2024
The power station consists of two identical gas-fired units with an initial capacity of 210 MW and a generating capacity of 180 MW.
- Kaptai Hydro Power Plant, Rangamati, Chittagong 2024
The only hydroelectric power plant in Bangladesh, utilizing water from the Karnaphuli River. It has five units with a total installed capacity of 230 MW.

Extracurricular Activities

- Volunteer, Projects For Humanity (**P4H**). Since 2026
Proud to contribute as an Assistant Teacher in the Education Projects of Projects for Humanity, helping empower the next generation through quality education.
- Volunteer, Education For All (**Guddi Foundation**). Since 2021
- Planning Secretary, Greater Mymensingh Association, CUET. 2021 - 2025
- Research Co-supervisor, CEMaterials and ML Lab, **ResearchBuddyAI**. 2025 - Present

Honors and Awards

- Regional Round Qualifier, Bangladesh Physics Olympiad 2018
- Dutch-Bangla Bank Scholarship(**DBBL**) 2019
- Guddi Foundation Scholarship 2019
- JSC Board Scholarship, Bangladesh 2014

References

Prof. Dr. Mahmud Abdul Matin Bhuiyan

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Prof. Dr. Kazi Delwar Hossain

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